RNA extraction (≥ 5 μg)

RNeasy® Plant Mini Kit (Manual) or RNeasy® Plus Mini Kit (Manual)

Preparation

Usage amount per sample:

Chemical	Concentration	Amount	Location
Buffer RLT mixture [α]	–	450 μL	RNA bench
Buffer RW1 [β]	–	700 μL	RNA bench
Buffer RPE working solution [γ]	–	1000 μL	RNA bench
DNAse I incubation mix [δ]	–	80 μL	fresh-prepared
Ethanol (EtOH)	≥ 96%	225 μL	V
RNase-free water	–	30 μL	RNA bench

[α] **Buffer RLT mixture**: Add 10 μL β-mercaptoethanol (β-ME) to 1 mL Buffer RLT. The mixture can be stored at room temperature for up to 1 month. Buffer RLT contains a high concentration of guanidine isothiocycanate, which supports the binding of RNA to the silica membrane. The β-mercaptoethanol should be added to Buffer RLT to inactivate RNases.

[β] **Buffer RW1** contains a guanidine salt, as well as ethanol, and is used as a stringent washing buffer that efficiently removes biomolecules such as carbohydrates, proteins, fatty acids etc., that are non-specifically bound to the silica membrane. At the same time, RNA molecules larger than 200 bases remain bound to the column.

[γ] **Buffer RPE working solution**: Add 4 volumes of ≥ 96% ethanol to Buffer RPE. In this case, add 2000 µL ethanol to 500 µL Buffer RPE. It is a mild washing buffer for removing traces of salts, which are still on the column due to buffers used earlier in the protocol.

[δ] **DNase I incubation mix**: Add **10 µL DNase I stock solution** to **70 µL Buffer RDD**, mix by inverting the tube, and put on ice.

> - **DNase I stock solution**: Located at 4°C (b). Dissolve the lyophilized DNase I in 550 µL RNase-free water. To avoid loss of DNase I, do not open the vial. Inject RNase-free water into the vial using an RNase-free needle and syringe. Mix gently by inverting the vial. Do not vortex. Store the stock solution as single use aliquots at −20°C (9 months) or 4°C (1 month). Do not refreeze after thawing.
> - **Buffer RDD**: Located at 4°C (b). It provides efficient on-column digestion of DNA and also ensures that the RNA remains bound to the column.

Procedure

All instruments must be sterilized prior to the experiment to ensure no RNA contamination and RNase residual.

Step 1: Cell lysis

1. Put in 5 beads in each 2 mL microcentrifuge tube. Add **450 µL Buffer RLT mixture** (contains 1% β-ME) into the eppendorf.
2. Dissect ≤ 100 mg tissue and immediately submerge them into the Buffer RLT mixture.

> - The rice root tip weight is approximately: [17.7, 20.6, 19.7, 21.2, 22.3] mg cm^{-1}/ 10 root-tips ≈ **0.2 mg / mm**. So the maximum is 250 root tips (cut 2 mm).
> - Put the tissue into an eppendorf, slightly centrifuge to spin down the tissue to the bottom of the eppendorf. If the tissue is not used immediately, store the sample at −80°C.

3. Disrupt the tissue using Tissue Grinder (**30 hits / sec**, **10 mins**).

> - Place a thick layer of tissue paper between the tissue grinder adapter cover and the eppendorf tube caps to boost grinding efficiency and prevent the tube caps from popping open during high-speed reciprocal shaking.

4. (optional) Water bath (**56°C**, **1 ~ 3 mins**). Help to disrupt the tissue and release RNA.
5. Now we get the **raw lysate**.

Step 2: Collect the lysate supernatant

1. Transfer the **raw lysate** to a **QIAshredder spin column** (purple lid) placed in a **2 mL collection tube**. Centrifuge (**full speed**, **2 mins**).

> - The QIAshredder spin column is used for removing cell debris and simultaneously homogenizes the lysate.

2. (optional) Transfer the supernatant to a **gDNA Eliminator Mini Spin Column** (colorless, with purple O-ring) placed in a 2 mL collection tube (supplied). Centrifuge (≥ **10,000 rpm** or 8000 × g, **30 secs**). Discard the column and save the flow-through.

> - The gDNA Eliminator Mini Spin Column is the component of RNeasy® Plus Mini Kit. It is not sold separately.

3. Transfer the supernatant of the flow-through to a new 1.5 mL microcentrifuge tube (not supplied) without disturbing the remaining cell-debris pellet.
4. Add **225 µL 96~100% ethanol** (≈ 0.5 volume) to the cleared lysate, and **mix well** immediately by pipetting. ***DO NOT centrifuge***. Proceed to the next step immediately.

> - The volume of lysate may be less than 450 µL due to loss during homogenization.
> - Precipitates may be visible after additon of ethanol. This does not affect the procedure.

Step 3: Bind RNA to column

1. Transfer the sample lysate (usually 650 µL), including any precipitate that may have formed, to an **RNeasy spin column** (pink lid) placed in a 2 mL collection tube (supplied). Close the lid gently.
2. Centrifuge (≥ **10,000 rpm** or 8000 × g, **15 secs**). **Discard the flow-through**. Reuse the collection tube.

> - If the sample volume exceeds 700 µL, centrifuge successive aliquots in the same RNeasy spin column. Discard the flow-through after each centrifugation.

Step 4: DNA digestion (optional)

1. Add **350 µL Buffer RW1** to the **RNeasy spin column**. Close the lid gently.
2. (optional) Incubate on benchtop (25°C, 3–5 mins) before centrifuging to enhance removal of excess gDNA prior to applying the enzyme.
3. Centrifuge (≥ **10,000 rpm** or 8000 × g, **15 secs**) to wash the **RNeasy spin column membrane**. **Discard the flow-through**. Reuse the collection tube.
4. Add **10 µL DNase I** to **70 µL Buffer RDD**. Mix well by gently inverting the tube. Centrifuge briefly to collect residual liquid from the sides of the tube.

> - DNase I is especially sensitive to physical denaturation. Mixing should only be carried out by gently inverting the tube. Do not vortex.

5. Add the **80 µL DNase I incubation mix** directly to the **RNeasy spin column membrane**, and place on the benchtop (**25 ± 5°C**, **15 mins**).

> - Be sure to add the DNase I incubation mix directly to the RNeasy spin column membrane. DNase digestion will be incomplete of part of the mix sticks to the walls or the O-ring of the spin column.

Step 5: Wash

1. Add **350 µL Buffer RW1** to the **RNeasy spin column**. Close the lid gently. Centrifuge (≥ **10,000 rpm** or 8000 × g, **15 secs**) to wash the spin column membrane. **Discard the flow-through**. Reuse the collection tube.
2. Add **500 µL Buffer RPE** to the **RNeasy spin column**. Close the lid gently. Centrifuge (≥ **10,000 rpm** or 8000 × g, **15 secs**) to wash the spin column membrane. **Discard the flow-through**. Reuse the collection tube.
3. Add **500 µL Buffer RPE** to the **RNeasy spin column**. Close the lid gently. Centrifuge (≥ **10,000 rpm** or 8000 × g, **2 mins**) to wash the spin column membrane. **Discard the collection tube with the flow-through**.

> - The long centrifugation dries the spin column membrane, ensuring that no ethanol is carried over during RNA elution. Residual ethanol may interfere with downstream reactions.
> - After centrifugation, carefully remove the RNeasy spin column from the collection tube so that the column does not contact the flow-through. Otherwise, carryover of ethanol will occur.

4. Place the **RNeasy spin column** in a **new 2 mL collection tube** (supplied). Close the lid gently. Centrifuge (**full speed**, **1 ~ 5 mins**).

> - Perform this step to eliminate any possible carryover of Buffer RPE, or if residual flow-through remains on the outside of the RNeasy spin column.

Step 6: Elute RNA

1. Place the **RNeasy spin column** in a **new 1.5 mL collection tube** (supplied).
2. Add 30–50 µL RNase-free water directly to the spin column membrane. Close the lid gently.

> - To increase RNA yield, incubate the spin column on the bench for 10 mins (optional).
> - If the expected RNA yield is >30 µg, repeat the elution step using another 30–50 µL RNase-free water. That means, you will get 60–100 µL eluate.

3. Centrifuge (≥ **10,000 rpm** or 8000 × g, **1 min**) to elute the RNA.

> - If high RNA concentration is required, repeat the elution step using the eluate. The RNA yield will be 15–30% less than that obtained using a second volume of RNase-free water, but the final RNA concentration will be higher.

4. Store the RNA eluate at −20°C (few weeks) or −80°C (> 1 year).

Done !!!
